Proceedings of the 6th International Conference on Pervasive Computing and Social Networking (ICPCSN-2026)
IEEE Xplore Part Number: CFP26EZ5-ART; ISBN: 979-8-3315-7236-5

2026 6th International Conference on Pervasive Computing and Social Networking (ICPCSN) | 979-8-3315-7236-5/26/$31.00 ©2026 IEEE | DOI: 10.1109/ICPCSN68523.2026.11543900

Improving Groundwater Quality Assessment through Gradient Boosting Approaches

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***Abstract*— Groundwater contamination is a problem because of factories and climate change. It affects the quality of water and peoples health. This study suggests using a computer program to predict groundwater quality. The program is called a Gradient Boosting Classifier. The program was trained on water data from 2018 to 2020. The data includes things like water chemistry and location. The new program was compared to programs like Random Forest and Multivariate Logistic Regression. The results show that the Gradient Boosting program is really good at predicting water quality with an accuracy of 96.30%. It is better than the programs. The Gradient Boosting program also shows precision, recall and F1-score. This means it can learn well and make predictions. The approach can help us understand and manage groundwater quality better. It supports making decisions about water resources. The Gradient Boosting Classifier is a tool, for predicting groundwater quality. Groundwater contamination is an issue that needs to be addressed. The Gradient Boosting program can help solve the groundwater contamination problem.**

Keywords—Groundwater quality, Gradient boosting classifier, Machine learning, Random forest, Water quality, Multivariate logistic regression, Groundwater contamination and Water resource.

I. INTRODUCTION

Groundwater is a source of fresh water for drinking, farming and industries especially in areas where surface water is scarce. However cities growing fast factories releasing waste and climate change have made groundwater quality worse making it hard to monitor and assess.[1] Traditional lab methods for checking groundwater quality are slow, expensive and not good for checking areas. Recently machine learning techniques like Random Forest, Support Vector Machine, Artificial Neural Networks and Gradient Boosting have been used to predict water quality. They are good at handling data.[2] Many studies have shown prediction performance, for example deep learning models have achieved R^2 values of up to 0.996, XGBoost has achieved an R^2 of 0.97, and Random Forest has achieved accuracy of up to 97%. Most studies far have used small datasets, looked at one parameter at a time, or classified water quality as just good or bad.[3] These limit their use in real-life situations. Also using methods like Gradient Boosting to classify groundwater quality into multiple classes with large datasets is still not explored much.[4] This study aims to create a model to predict groundwater quality using a Gradient Boosting Classifier. The model uses a -year dataset with many parameters to improve accuracy. Its performance is compared with methods like Random Forest and Multivariate Logistic Regression. The main contribution of this work is creating a high-accuracy model (96.30%) that helps make predictions and supports decisions for managing groundwater resources sustainably.[5] The model uses Gradient Boosting Classifier to classify groundwater quality. Gradient Boosting Classifier is used to improve prediction accuracy. The accuracy of Gradient Boosting Classifier is compared with Random Forest and Multivariate Logistic Regression. The results show that Gradient Boosting Classifier achieves accuracy.[6] Groundwater quality prediction is important, for groundwater resource management. Groundwater resource management relies on predictions. Accurate

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predictions support data-driven decision-making. Data-driven decision-making ensures use of groundwater resources [7].

II. RELATED WORKS

The total number of articles published on this topic over the last five years is more than 310 papers in IEEE Xplore, 212 papers in Google Scholar, and 152 papers in academia.edu.

The catboost applied in this model shows the groundwater depth over 43,000 km² at 10 m resolution using 13,000 wells and 19,000 proxy observations. The model produced an average error of close to 115 cm with a further reduction to less than 50 cm when proxy data were added. [8]. To predict groundwater nitrate concentration in Iran, their outputs were connected using geographically weighted regression (GWR) to improve the accuracy. The findings demonstrated that the ensemble of ML models using GWR resulted in the highest performance (R2 is 0.75 and RMSE is 9.38 mg/l) compared to an ensemble model [9]. Using 2000-2013 data. For the Rayee well, the RRMSE values for GTB, GMDH, and ANFIS were 0.473, 0.517, and 0.7522, respectively. [10]. The EPR model achieved the best performance with a correlation coefficient R is 0.9999 and RMSE is 0.2105, outperforming MARS, M5MT, and GEP. Based on the MARS model, the region was classified into 73.06% very low and 26.94% low risk zones. [11]. A Principal component regression based regression model gained 95% prediction accuracy for water quality assessment and Gradient Boosting model is 100% accuracy, This beating existing methods [12]. An groundwater potential estimate RF, Bagged CART, AdaBoost, and GamBoost using 339 groundwater areas and 15 conditioning factors and mean accuracy is 0.84. The Random Forest model performed best with accuracy is 0.86, Kappa is 0.67, precision is 0.85, and recall is 0.91. [13]. The Logistic Regression, KNN, SVM variants, Decision Tree, AdaBoost, and XGBoost were evaluated. The results show AdaBoost, XGBoost, and Polynomial SVM achieved higher accuracy ranging from 92% to 98% [14]. A study on groundwater contamination prediction used Multivariate logistic regression and Random forest on 11508 samples with ammonia as a contamination indicator. The Random Forest model is 93% accuracy and Multivariate logistic regression performance improved from 70% to 83% [15].

The proposed method used is Gradient Boosting algorithm to improve the groundwater quality by analyzing the different patterns. Compare both Random Forest and Multivariate Logistic Regression methods with Gradient Boosting show high performance. The input parameters used are Spatial / Location Parameters and Physicochemical Water Quality Parameters.

III. MATERIALS AND METHODS

The process was carried out in the MCA Lab at KSR College of Engineering. The dataset was gathered from Kaggle.com named groundwater quality dataset period from 2018 to 2020. The dataset includes parameters such as pH, (EC), (TDS), major ions, geographical attributes and more. Using Gradient Boosting Classifiers and in Python the scikit- learn library for predicting water quality. The model was trained and evaluated using train and test splitting and cross- validation [16].

Group 1 The traditional ML model uses both Multivariate logistic regression and Random forest algorithm. The dataset contains over 11508 samples that are used to predict groundwater quality and also this predictive model uses 9 input parameters [17]. Group 2 The modern model implements the Gradient Boosting Classifiers and the dataset contains around 11508 samples and 12 input parameters which make this model more accurate prediction in ground water quality. The main Target Variables are Groundwater Quality Class like C2S1, C3S1, C3S2, C4S1 and so on.

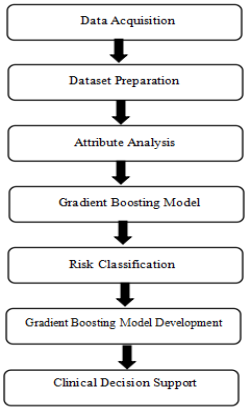


Fig. 1. Workflow of Predictive analytics algorithm Model

Figure 1. Starting from data acquisition. The three-year groundwater quality data collected from monitoring sources are collected. In the dataset preparation stage, several processing steps clean, pre-process, and organize the data, followed by attribute analysis in the search for important physicochemical and spatial features that enhance the accuracy of the model in predicting groundwater quality. A Gradient Boosting model is developed and trained to learn complex patterns and then tested; it performs risk classification to divide the groundwater into several classes of contamination. After that, the model so trained gives decision support to authorities for informed choices in the assessment and management of groundwater quality.

VI. STATISTICAL ANALYSIS

SPSS is used for advanced statistical analysis. The tool is used to compare the proposed and existing method using independent samples [18]. The dependent variable is the groundwater quality

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classification and independent variables are physicochemical, hydrogeological, spatial, and temporal groundwater parameters.

V. RESULT

The Multivariate Logistic Regression (MLR), Random Forest (RF), and the proposed model Gradient Boosting Classifier, has better performance against the existing models. Analysis used the t-test statistical significance of the performance compared to the existing methods. Table 1 The Gradient Boosting Classifier accuracy, precision, and recall is 96.30%, 98.00%, and 96.00% achieved, and the RF model the accuracy is 93%, precision 93%, and recall 92%, while MLR achieved accuracy 83%, precision 91%, and recall 88%. Table 2 the improvement of mean accuracy of Gradient Boosting Classifier is 96.30% across the mean accuracy of RF 93.00% and MLR 83.00%, showing a good improvement. The SD for Gradient Boosting Classifier is 0.041, which is lower than RF 0.054 and MLR 0.081, indicating good achieved. Fig. 2 the accuracy graph shows 96.30% achieved by Gradient Boosting Classifier compared to Random Forest 93% and Multivariate Logistic Regression 83%. It highlights the learning capability of Gradient Boosting Classifiers. Fig. 3 Represent precision comparison between the MLR, RF, and Gradient Boosting Classifier models. The Gradient Boosting Classifier model achieved higher precision 0.98 compared to the RF 0.93 and MLR 0.91. Difference shows the proposed method is classified better than the existing methods. Fig. 4 Represent F1-score comparison between the MLR, RF, and Gradient Boosting Classifier models. The Gradient Boosting Classifier model achieved higher F1-score 0.96 compared to the Random Forest 0.91 and Multivariate Logistic Regression 0.89. Difference shows the proposed method is classified better than the existing methods.. Fig. 5 Represent recall comparison between the MLR, RF and Gradient Boosting Classifier models. The Gradient Boosting Classifier model achieved higher recall 0.96 compared to the Random Forest 0.92 and Multivariate Logistic Regression 0.88. Difference shows the proposed method is classified better than the existing methods. Fig. 6 The Gradient Boosting Classifier with accuracy 96.30%, precision 98.00%, recall 96.00%, and F1-score 96.00%, and the t-test confirms the Gradient Boosting Classifier model's superior performance ($p < 0.05$).

Gradient Boosting Classifier achieved better performanceagainst the Multivariate Logistic Regression and Random Forest model not only in the accuracy but also in stability, reliability and overall consistency and effective model.

Comparative Analysis of Machine Learning Models

The following table presents a comparative analysis of the performance of the proposed Gradient Boosting Classifier with existing models such as Random Forest and Multivariate Logistic Regression based on accuracy, precision, and recall metrics.

Table 1: Comparison of Machine Learning Models based on Performance Metrics

Model	Accuracy (%)	Precision	Recall
Multivariate Logistic Regression (MLR)	83	0.91	0.88
Random Forest (RF)	93	0.93	0.92
Gradient Boosting Classifier (Proposed)	**96.30**	**0.98**	**0.96**

TABLE 2 ACCURACY, PRECISION, AND RECALL USED FOR OVER 20 SAMPLES. THE IMPROVEMENT OF 13.30% ACCURACY ACHIEVED WHILE COMPARED THE EXISTING MODEL MLR (83%) WITH THE RF (93%) AND THE PROPOSED MODEL GRADIENT BOOSTING CLASSIFIER (96.30%), THE SAME PATTERN IS FOLLOWED IN BOTH PRECISION AND RECALL, VALUES IMPROVED FROM 0.88 TO 0.92 AND FINALLY TO 0.96. GRADIENT BOOSTING CLASSIFIERS HAVE THE BEST LEARNING CAPABILITY.

S.No	Sample	RF Accuracy(%)	GradientBoostingAccuracy(%)	MLR Precision(%)	GradientBoostingPrecision(%)	MLR Recall(%)	GradientBoostingRecall(%)
1	Sample1	91.2	95.8	78.5	94.6	80.1	95.2
2	Sample2	92.0	96.1	79.0	95.0	81.0	95.8
3	Sample3	92.5	96.3	80.2	95.1	82.4	96.0
4	Sample4	93.0	96.5	81.0	95.3	83.0	96.1
5	Sample5	91.8	95.9	79.4	94.8	81.6	95.5

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6	Sample6	92.6	96.4	80.8	95.2	82.9	96.0
7	Sample7	93.1	96.7	81.6	95.4	83.7	96.3
8	Sample8	92.3	96.2	80.0	95.0	82.1	95.9
9	Sample9	91.9	95.8	79.1	94.7	81.2	95.4
10	Sample 10	92.7	96.4	80.9	95.2	83.0	96.1
11	Sample11	93.0	96.6	81.3	95.3	83.5	96.2
12	Sample12	92.1	96.0	79.6	94.9	81.8	95.6
13	Sample13	92.8	96.5	81.0	95.4	83.2	96.2
14	Sample14	93.2	96.8	82.0	95.6	84.1	96.4
15	Sample15	92.4	96.2	80.4	95.1	82.6	95.9
16	Sample16	91.7	95.7	78.9	94.6	81.0	95.3
17	Sample17	92.9	96.6	81.5	95.5	83.6	96.3
18	Sample18	93.0	96.7	81.8	95.6	83.9	96.4
19	Sample19	92.2	96.1	80.1	95.0	82.3	95.8
20	Sample20	93.0	96.3	81.9	95.5	84.0	96.3

TABLE 3 MULTIVARIATE LOGISTIC REGRESSION, AND THE STANDARD DEVIATION IS 0.041 IS ACHIEVED FROM GRADIENT BOOSTING CLASSIFIER, 0.054 FROM RANDOM FOREST, AND 0.081 FROM MULTIVARIATE LOGISTIC REGRESSION, AND STANDARD ERROR MEAN IS 0.009 IS LOWER COMPARED TO 0.012 AND 0.018.

Model	N (Samples)	Mean Accuracy	Std. Deviation	Std.Error Mean
MLR	20	83.00%	0.08120	0.01815
Random Forest	20	93.00%	0.05460	0.01221
Gradient Boosting Classifier	20	96.30%	0.04185	0.00936

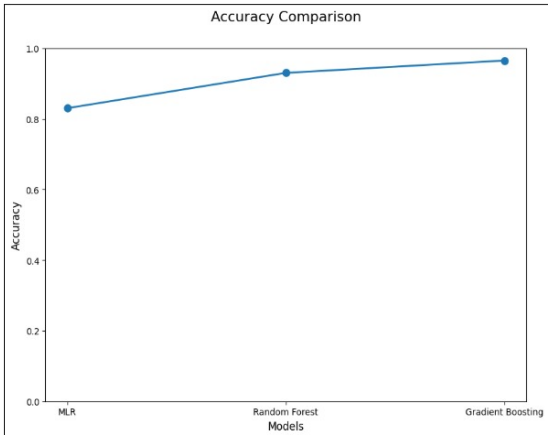


FIG. 2 THE ACCURACY COMPARISON BETWEEN THE MLR, RF,AND GRADIENT BOOSTING CLASSIFIER MODELS. THE

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GRADIENT BOOSTING CLASSIFIER MODEL ACHIEVED HIGHER ACCURACY 96.30% COMPARED TO THE RF 93% AND MLR 83%. DIFFERENCE SHOWS THE PROPOSED METHOD IS CLASSIFIED BETTER THAN THE EXISTING METHODS.

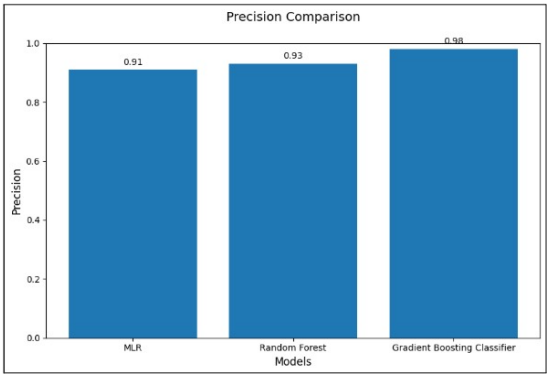


FIG. 3 THE PRECISION COMPARISON BETWEEN THE MLR, RF, AND GRADIENT BOOSTING CLASSIFIER MODELS. THE GRADIENT BOOSTING CLASSIFIER MODEL ACHIEVED HIGHER PRECISION 0.98 COMPARED TO THE RF 0.93 AND MLR 0.91. DIFFERENCE SHOWS THE PROPOSED METHOD IS CLASSIFIED BETTER THAN THE EXISTING METHODS.

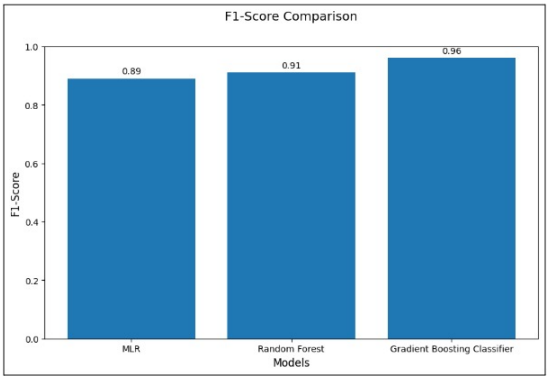


FIG. 4 THE F1-SCORE COMPARISON BETWEEN THE MLR, RF, AND GRADIENT BOOSTING CLASSIFIER MODELS. THE GRADIENT BOOSTING CLASSIFIER MODEL ACHIEVED HIGHER F1-SCORE 0.96 COMPARED TO THE RANDOM FOREST 0.91 AND MULTIVARIATE LOGISTIC REGRESSION 0.89. DIFFERENCE SHOWS THE PROPOSED METHOD IS CLASSIFIED BETTER THAN THE EXISTING METHODS.

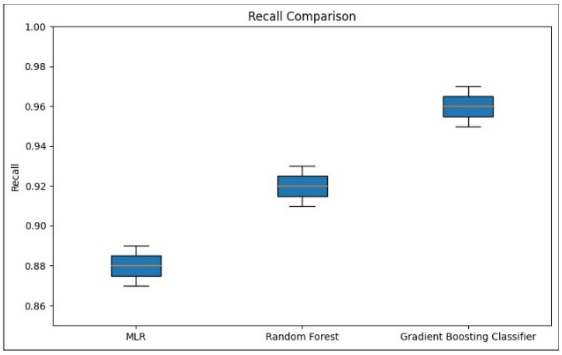


FIG. 5 THE RECALL COMPARISON BETWEEN THE MLR, RF AND GRADIENT BOOSTING CLASSIFIER MODELS. THE GRADIENT BOOSTING CLASSIFIER MODEL ACHIEVED HIGHER RECALL 0.96 COMPARED TO THE RANDOM FOREST 0.92 AND MULTIVARIATE LOGISTIC REGRESSION 0.88. DIFFERENCE SHOWS THE PROPOSED METHOD IS CLASSIFIED BETTER THAN THE EXISTING METHODS.

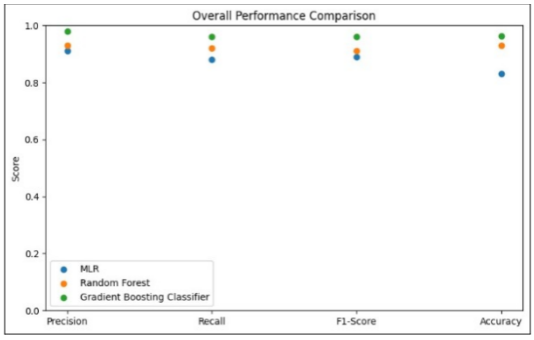


FIG. 6 THE GRADIENT BOOSTING CLASSIFIER OUTPERFORMS IN THE FOUR METRICS, PRECISION 98.00% AND SCORE IS 0.9800, RECALL 96.00% AND SCORE IS 0.9600, F1 IS 96.00% AND SCORE 0.9600, ACCURACY 96.30% AND SCORE IS 0.9630. THE GRAPH COMPARISON SHOWS THE GRADIENT BOOSTING CLASSIFIER IS MORE SUITABLE WHILE COMPARED TO THE RANDOM FOREST AND MULTIVARIATE LOGISTIC REGRESSION MODELS.

VI. DISCUSSION

Discussing ground water quality prediction using the Gradient Boosting algorithm, Achieved a statistically significant prediction accuracy of 96.30% which shows a strong prediction in water quality. By using the testing sample, give weighted precision of 0.95, recall of 0.96, and F1-score of 0.96,

Using several ML algorithms such as the random Forest, the parallel random Forest, the extreme gradient boosting (XGB) using various boosters. PRF and XGB (linear booster) obtained around 86% accuracy on the basis of AUC (ROC analysis) [19]. Two different models are combining Gradient Boosting and Ridge Regression, this method called Stacking Regressor. By using that the Model calculates Mean Squared Error (MSE) is 0.012 and R-squared (R^2) is 0.94 [20]. With

248 groundwater monitoring wells the study area covers Unconfined aquifers in northern Iran. By using this data the model uses a different algorithm. Mainly forest achieve the best performance Overall Accuracy is 0.92, Precision is 0.92, Recall is 0.92, ROC is 0.95 [21]. Model almost 97 watersheds data and, The main region is Southeast Atlantic, USA and 28 climate and catchment-related variables, Both Random Forest and Boosted Regression tree are used in this model and the random forest model achieve more accuracy than boosted regression tree TN and TP increased seriously when urban and agricultural land exceeded 40% and 43% [22].

Even though it contains three years of dataset, it only has a limited datasets with small sample sizes, which make the model low robust and scalable. Only focused on single parameter or Binary classification, so it failed to address the multiple parameter classification of groundwater contamination. By using the large scale dataset and Multiclassification, we can make morning more strong, and also we can move groundwater condemnation to deep learning, which makes it more accurate in predicting groundwater quality [23].

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VII. CONCLUSION

The article was developed to predict a model of groundwater contamination based on Gradient Boosting and combine multi-year groundwater quality data and various physicochemical and spatial parameters. High accuracy of 96.30% with high values of accuracy, precision, and F1-score, thus can be used in real-world applications. Moreover, it has been observed under comparative analysis that the developed Gradient Boosting methodology is better than the traditional machine learning models.

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